

Hojeong Lee

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EDUCATION

- Korea Advanced Institute of Science and Technology** | Ph.D. Candidate
School of Electrical Engineering, Robotics Program (Advisor: Sang Ho Yoon) Daejeon, Korea | Aug 2025 -
- Korea Advanced Institute of Science and Technology** | Master of Science
Graduate School of Culture Technology (Advisor: Sang Ho Yoon) Daejeon, Korea | Aug 2023 - 2025
- Thesis: 'Wearable Haptic Interface for Shape Perception through Spatiotemporal Vibrotactile Patterns and Kinesthetic Feedback'
- Seoul National University** | B.F.A in Industrial Design
Seoul, Korea | Mar 2018 - Aug 2023

PUBLICATION

Journals and Letters

- **EStatiG: Wearable Haptic Feedback with Multi-Phalanx Electrostatic Brake for Enhanced Object Perception in VR**
Nicha Vanichvoranun*, **Hojeong Lee***, Seoyeon Kim, Sang Ho Yoon (* **First Co-author**)
Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT), Vol. 8, No. 3, Article 131. (2024.9) (IF 4.5, Q1)
- **Spatiotemporal Vibration Based Multimodal Haptic Rendering with a Lightweight Exo-Glove for 3D Shape Perception**
Hojeong Lee, Eunho Kim, Rachel Kim, Sang Ho Yoon
IEEE Transactions on Haptics (ToH) (**Under Review**)
- **Exoskeleton for 2-DoF kinesthetic feedback**
Daegeun Park, **Hojeong Lee**, Sang Ho Yoon, Sung-uk Jung
IEEE Robotics and Automation Letters (RA-L) (**Under Review**)

Conference Papers

- **3D Shape Perception through Spatiotemporal Vibrotactile Patterns with Kinesthetic Feedback**
Hojeong Lee, Eunho Kim, Sang Ho Yoon
Proc. IEEE World Haptics Conference (WHC, 2025.7)
- **Optimizing Button Size to Enhance User Experience for Augmented Reality Interactions in Mobile Scenarios**
Hojeong Lee, Yeahyoung Jeon, Keisuke Kudo, Paul Ssemakula, Seonghyeok Park and Shuping Xiong
Proc. 22nd Triennial Congress of International Ergonomics Association (IEA, 2024.8)
- **ThermicVib: Enabling Dynamic Thermal Sensation with Multimodal Haptic Glove for Thermal-Responsive Interaction**
Hyungil Yi, **Hojeong Lee**, Sang Ho Yoon
Proc. IEEE International Symposium on Mixed and Augmented Reality (ISMAR, 2024.10)

Posters and Demo

- **Effect of Movement Velocity and Flexion Angle on Realistic Haptic Reproduction**
Dongkyu Kwak*, **Hojeong Lee***, Sang Ho Yoon (***First Co-Author**)
IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR Adjunct, 2025.10)
- **Adaptive and Immersive XR Interactions with Wearable Interfaces (Demo of KAIST HCI Tech Lab)**
Sang Ho Yoon, Youjin Sung, Kun Woo Song, Kyungeun Jung, Kyungjin Seo, Jina Kim, Hyungil Yi, Nicha Vanichvoranun, Hanseok Jeong, **Hojeong Lee**
Proc. 2024 CHI Conference on Human Factors in Computing Systems (CHI Interactivity, 2024.5)
(**People's Choice Best Demo Award**)

HONORS AND AWARDS

Dean's Research Award College of Liberal Arts and Convergence Science, KAIST	2025
Popular Choice Winner CHI'24 Interactivity	2024

RESEARCH PROJECTS

Development of a multimodal haptic rendering technology based on atypical objects Electronics and Telecommunications Research Institute (ETRI)	Jan 2024 -
Emotional Expression through Robot Arm Movement Co-work with XYZ Corporation	Aug 2025 -
A wearable multimodal sensing framework for adaptive interaction in extended reality National Research Foundation (NRF)	Aug 2023 - Aug 2025
Developing the model and criteria of evaluating motion sickness in automobile driving scenarios Hyundai NGV - Human Interface System Lab (Professor Myung Hwan Yun, Seoul National University)	Jan 2023 - Jun 2023

EXPERIENCE

Frankfurt University of Applied Science Summer School Exchange Student - Urban Design and Architecture Program, "Urban Redesign Project for Frankfurt Neue Strasse and Offenbach Station"	Frankfurt, Germany July 2019
Student President of the Department of Design Seoul National University	Dec 2019 - Dec 2020

INVITED TALKS

Invited Talk at Seoul National University, SNU Design Week Title: "From the Designer and Artist to HCI Researcher"	Dec 2025
Invited Talk at Korea Haptics Conference Title: "Wearable Haptic Feedback with Multi-Phalanx Electrostatic Brake for Enhanced Object Perception in VR"	Aug 2024

ACADEMIC SERVICES

Review CHI LBW 24', Augmented Humans 25', WHC 25', ISMAR 25', UIST 25', ISMAR Poster 25', CHI 26', HRI 26'
Student Volunteer UbiComp/ ISWC 2024

EXHIBITIONS

Seoul Design 2022 (2022, Granted by Seoul Design Foundation 서울디자인재단) 매몽매몽 Dream Trade (Haeun Ki, Yuri Kim, Heejung Yu, Hojeong Lee, Seyun Jun, Sungyun Jang) Designer (Directing, 3D design, Mock-up, Web Design)	Oct 2022
ZER01NE DAY 2021 (2021, Granted by Hyundai Motors Group) headspace (Keunwook Kim) Designer Assistant (3D design, Exhibition Design, Prototyping)	Nov 2021
Artspace@SNU (Funded by Seoul National University) 원 Wishes (Hojeong Lee) Designer (Directing, Interaction Design, Visual Design)	Oct 2021